

Tulga-Erdene Sodjargal

✉ tulgaerdene.sodjargal@gmail.com | [tulga-rdn.github.io](https://github.com/tulga-rdn)

[in](#) [tulga-erdene](#) | [github](#) [tulga-rdn](#) | [Google Scholar](#)

EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)** Aug. 2021 – Feb. 2027 (expected)
B.S. in Bio and Brain Engineering, Double Major in Chemistry Daejeon, South Korea
 - Cumulative GPA: 92.3/100 (as of Sep. 2026).
 - **ML & Statistics:** Statistical ML, Biomedical Statistics and ML, Big Data and ML in Biotechnology, Probability and Statistics.
 - **Chemistry & Computation:** ML for Molecules and Materials (graduate), AI Chemistry, Computational Chemistry, Physical Chemistry I–II (quantum chemistry, statistical thermodynamics).
- **École Polytechnique Fédérale de Lausanne (EPFL)** Sep. 2024 – Jun. 2025
Visiting & Project Student Lausanne, Switzerland
 - Exchange semester with part-time research, followed by full-time research project.

PUBLICATIONS & PRESENTATIONS

J=JOURNAL, S=IN SUBMISSION, P=POSTER

- [J1] E Rumiantsev, MF Langer, **T-E Sodjargal**, M Ceriotti, P Loche (2026). **Learning Long-Range Representations with Equivariant Messages**. *Transactions on Machine Learning Research (TMLR)*.
- [S1] A Goscinski, TJ Baird, D Du, J Prado, D Suman, **T-E Sodjargal**, S Bonella, G Pizzi, M Ceriotti (2025). **scicode-widgets: Bringing Computational Experiments to the Classroom with Jupyter Widgets**. Under review at *Journal of Open Source Education (JOSE)*. arXiv:2507.05734.
- [P1] **T-E Sodjargal**, E Rumiantsev, P Loche, M Ceriotti (2025). **Integrating Long-Range Interactions into Machine Learning Interatomic Potentials**. Poster, *German Physical Society (DPG) Spring Meeting*, session MM 9.62.

RESEARCH EXPERIENCE

- **Data Science & Artificial Intelligence Lab, KAIST (Prof. Chanyoung Park)** Jun. 2026 – Present
Research Intern (LLMs for Materials Design) Daejeon, South Korea
 - Re-architected an LLM-driven materials-design pipeline, replacing evolutionary search with reinforcement learning.
 - Implemented chemically valid tools for programmatic modification and characterization of material structures.
 - **Tools:** PyTorch, OpenAI API
- **University of Illinois Urbana-Champaign (Prof. Taras V. Pogorelov)** Feb. 2024 – Present
Research Intern (Molecular Dynamics) Remote
 - Analyze all-atom molecular-dynamics trajectories of cell membranes with varying lipid compositions for multi-team drug-design projects.
 - Refactored and vectorized the analysis pipeline, cutting runtime by $> 30\times$ and making previously intractable structural analyses feasible.
 - Co-authoring a manuscript (in preparation).
 - **Tools:** MDAnalysis, scikit-learn, NumPy
- **Laboratory of Computational Science and Modeling, EPFL (Prof. Michele Ceriotti)** Sep. 2024 – Sep. 2025
Research Intern (Atomistic ML) Lausanne, Switzerland
 - Developed ML interatomic potentials with long-range interactions, improving accuracy for molecular property prediction and MD.
 - Co-authored a *TMLR* paper and presented first-author results as a poster at the German Physical Society (DPG) Spring Meeting 2025.
 - Contributed to open-source atomistic-ML libraries (PyTorch optimizations, CI/CD) and interactive Jupyter teaching materials.
 - **Tools:** PyTorch, ASE, LAMMPS, ipywidgets, Git

- **Wellman Center for Photomedicine, Harvard Medical School (Prof. Mei X. Wu)** Jun. 2025 – Aug. 2025
Research Intern (Computational Biochemistry) Boston, USA
 - Established the group's computational research capability as the sole dry-lab researcher in an otherwise wet-lab team.
 - Independently designed and ran molecular-docking and simulation studies to rationalize and guide experimental results.
 - **Tools:** AutoDock Vina, NAMD, Gaussian
- **Therapeutic Protein Design and Structural Biology Lab, KAIST (Prof. Byung-Ha Oh)** Jun. 2023 – Nov. 2023
Undergraduate Researcher (Protein Design) Daejeon, South Korea
 - Designed monomeric Fc variants and FcRn-binding proteins with a computational pipeline: RFdiffusion backbone generation, ProteinMPNN sequence design, and AlphaFold2 structure validation.
 - Automated the workflows with Bash scripts.
 - **Tools:** RFdiffusion, ProteinMPNN, AlphaFold2, Bash
- **Sustainable Catalysis Lab, KAIST (Prof. Yoonsu Park)** Mar. 2022 – Jun. 2023
Undergraduate Researcher (Synthetic Chemistry) Daejeon, South Korea
 - Diversified the catalyst library by exploring new catalyst scaffolds for combinatorial reaction screening.
 - Ran air- and moisture-sensitive organometallic reactions using Schlenk-line and glovebox techniques; characterized products by 1D and 2D NMR.

INDUSTRY EXPERIENCE

- **PuzzleAI** Jan. 2026 – Feb. 2026
Research Intern (LLM Data Engineering) Daejeon, South Korea
 - Built a pipeline for LLM-driven expansion of single-turn datasets, with automated Korean-to-English translation and I/O parsing.
 - **Tools:** Python, OpenAI API
- **SpiderCore Inc.** Jun. 2024 – Aug. 2024
Research Intern, ML for Molecules (B.S. Thesis) Daejeon, South Korea
 - Built graph neural networks for gene-therapy design, bringing chemical domain knowledge into model architecture choices.
 - Devised a chemically inspired self-supervised pretraining task that raised model performance to state-of-the-art levels.
 - **Tools:** TensorFlow, RDKit

SKILLS

- **Programming:** Python, R, MATLAB, Bash, SQL, Git, CI/CD, pytest
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Optuna, OpenAI API
- **Molecular Modeling:** ASE, LAMMPS, NAMD, Gaussian, AutoDock Vina, MDAnalysis, RDKit, RFdiffusion, ProteinMPNN, AlphaFold2
- **Wet Lab:** Schlenk line and glovebox techniques, flash chromatography, $^1\text{H}/^{13}\text{C}$ NMR (1D, 2D)
- **Languages:** English (TOEFL iBT 116/120), Russian (native), Mongolian (native), Korean (basic)

SCHOLARSHIPS & AWARDS

- **Research Internship Funding** 2025
KAIST; 3000 USD for covering program fees for summer research at Harvard Medical School
- **Exchange Scholarship** 2024
KAIST; 3300 USD for the exchange semester at EPFL
- **Silver Award** 2023
5th Place among teams from leading Korean STEM universities for building cost-aware demand forecasting models
- **Full Undergraduate Scholarship** 2021
KAIST; 4 years of tuition, monthly stipend, and health insurance